Physiology and function of Hb & anemia

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Introduction

- The main function of red blood cell
  - Transfer of O_2 from lungs to tissue
  - Transfer of CO_2 from tissue to lungs
- To accomplish this function red cells has haemoglobin (Hb)
- Each red cell has 640 million molecules of Hb

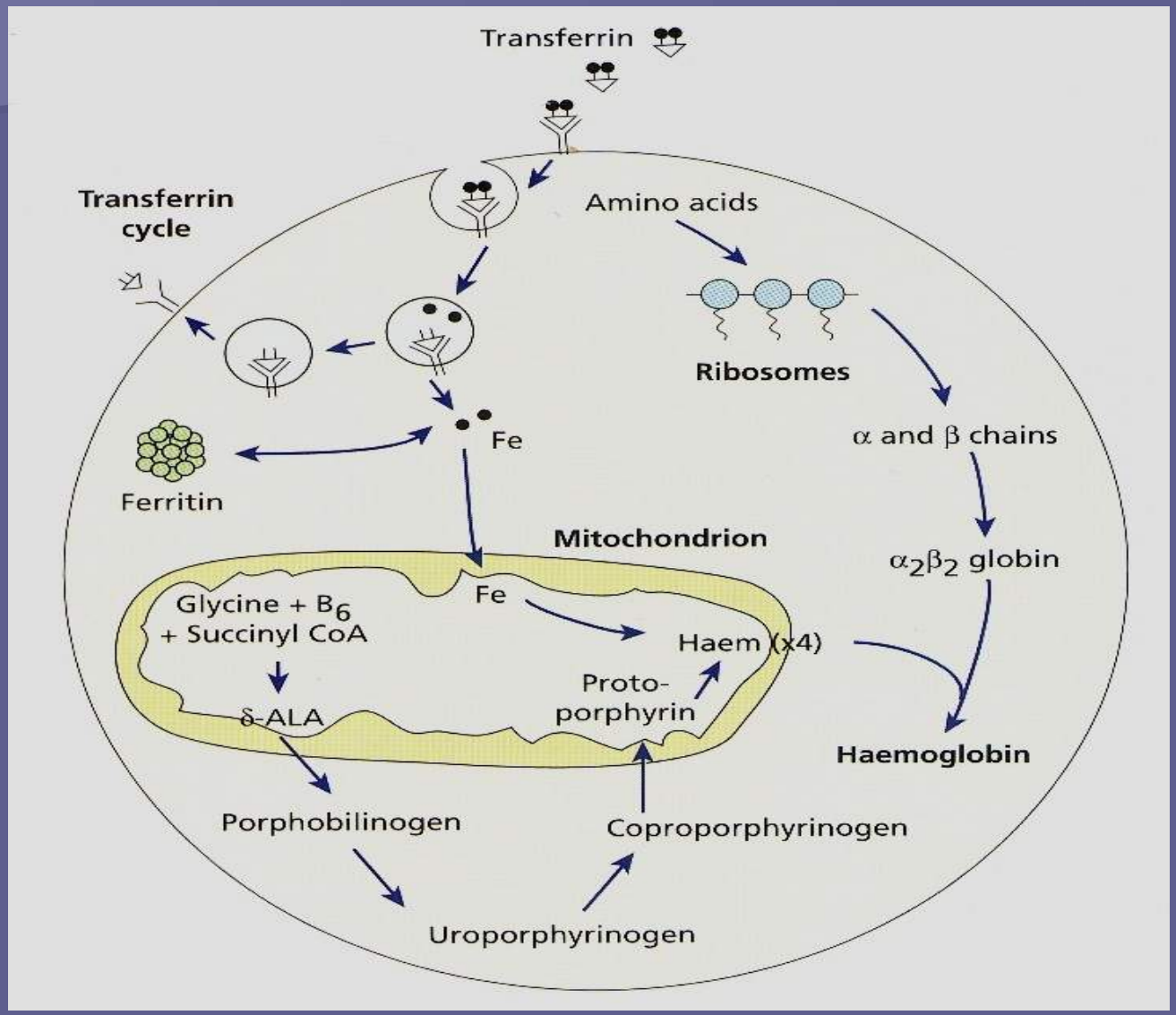
Introduction

- Haemoglobin (Hb), protein constituting 1/3 of the red blood cells
- Synthesis begins in proerythroblast
  - 65% at erythroblast stage
  - 35% at reticulocyte stage
- Two parts
  - Haem
  - Globin

Synthesis of Haemoglobin (Hb)

- Haem & globin produced at two different sites in the cells
  - Haem in mitochondria
  - Globin in polyribosomes
- Well synchronized

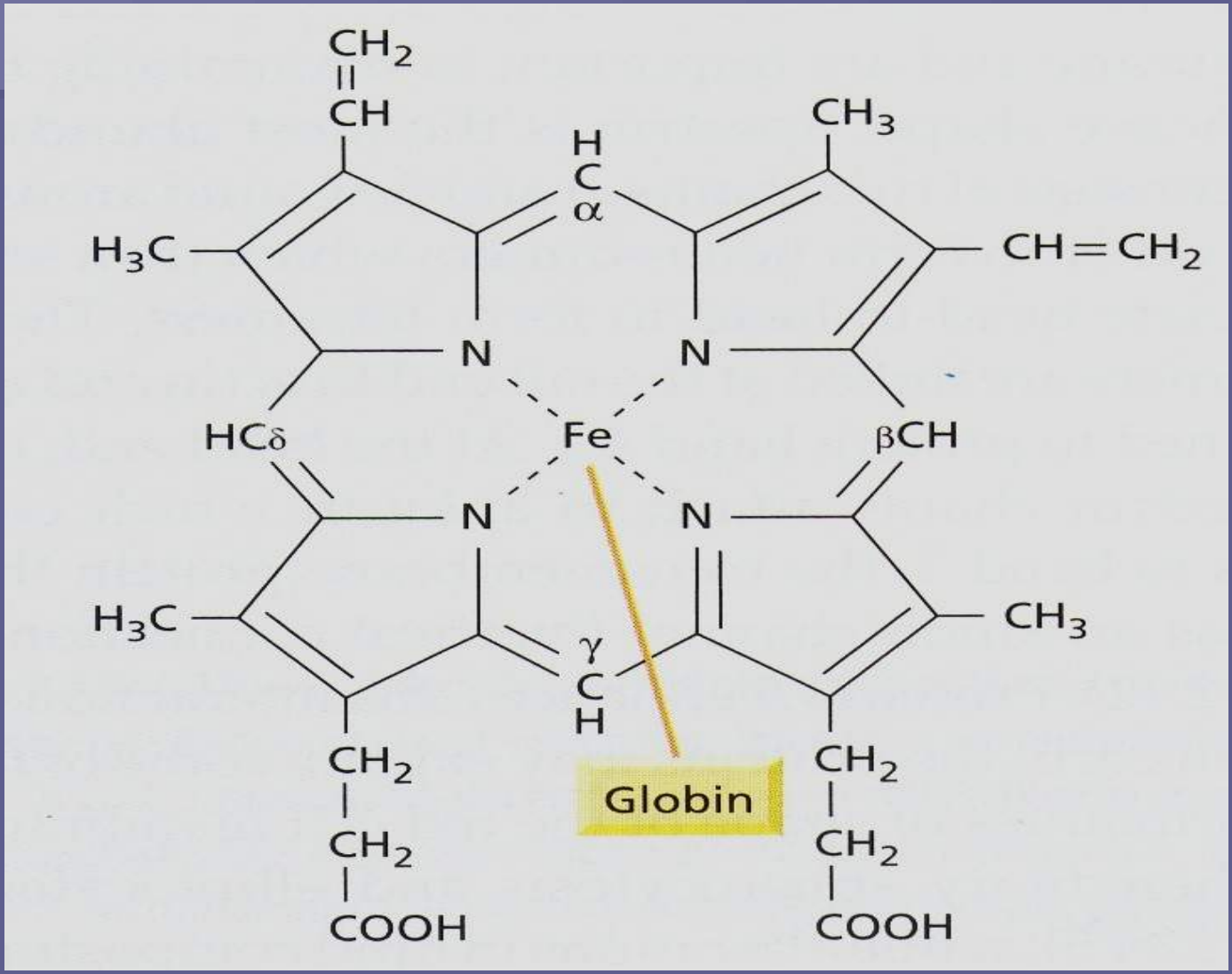
Synthesis of Haemoglobin



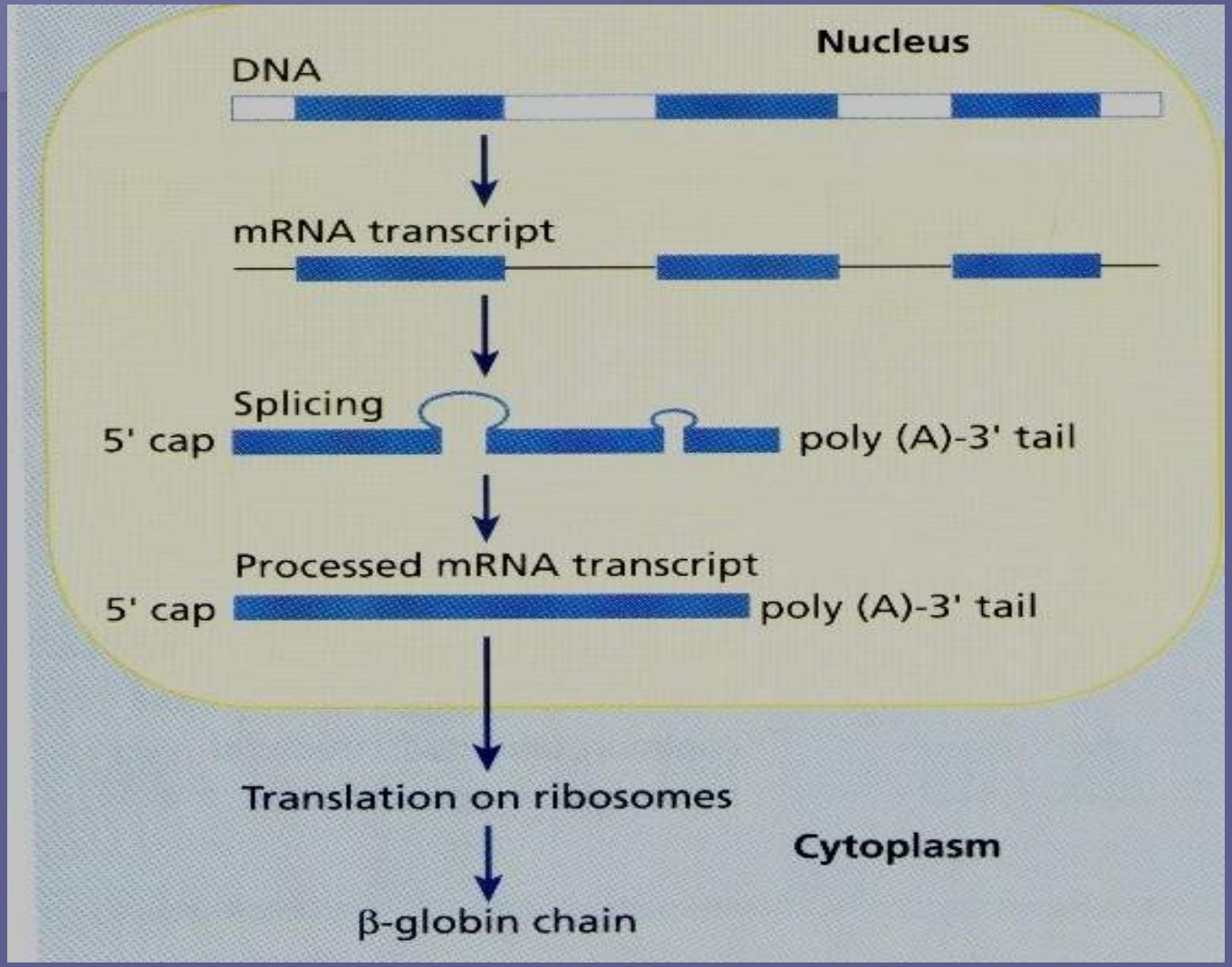
Synthesis of Haem

- Protoporphyrin ring with an iron atom in centre
- The main site is mitochondria
- Mature red cell does not contain mitochondria

Structure of Haem



Synthesis of globin

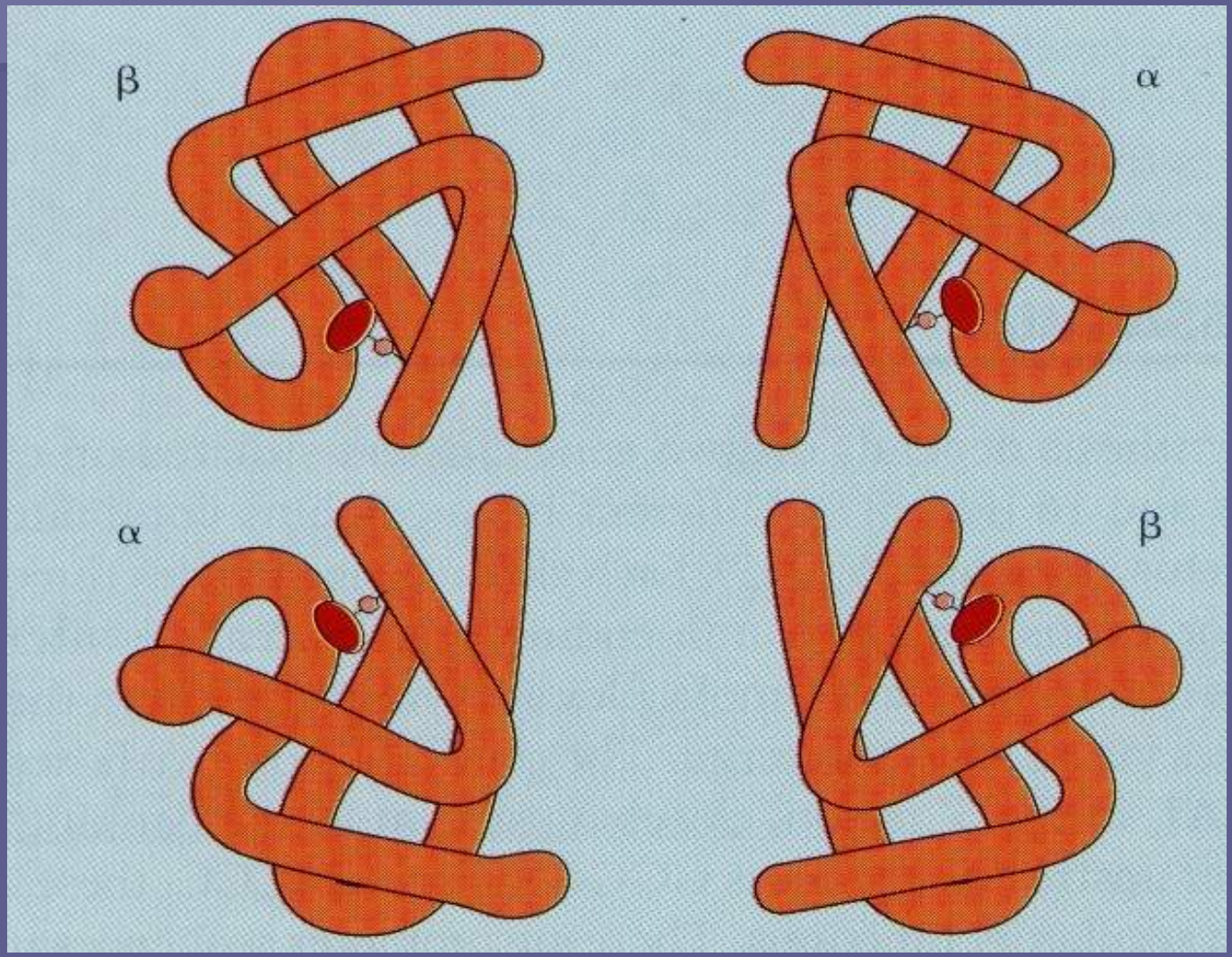


Synthesis of globin

- Various types of globin combines with haem to from different haemoglobin

- Eight functional globin chains, arranged in two clusters the
  - β- cluster (β, γ, δ and ε globin genes) on the short arm of chromosome 11
  - α- cluster (α and ζ globin genes) on the short arm of chromosome 16

Alpha & beta chains



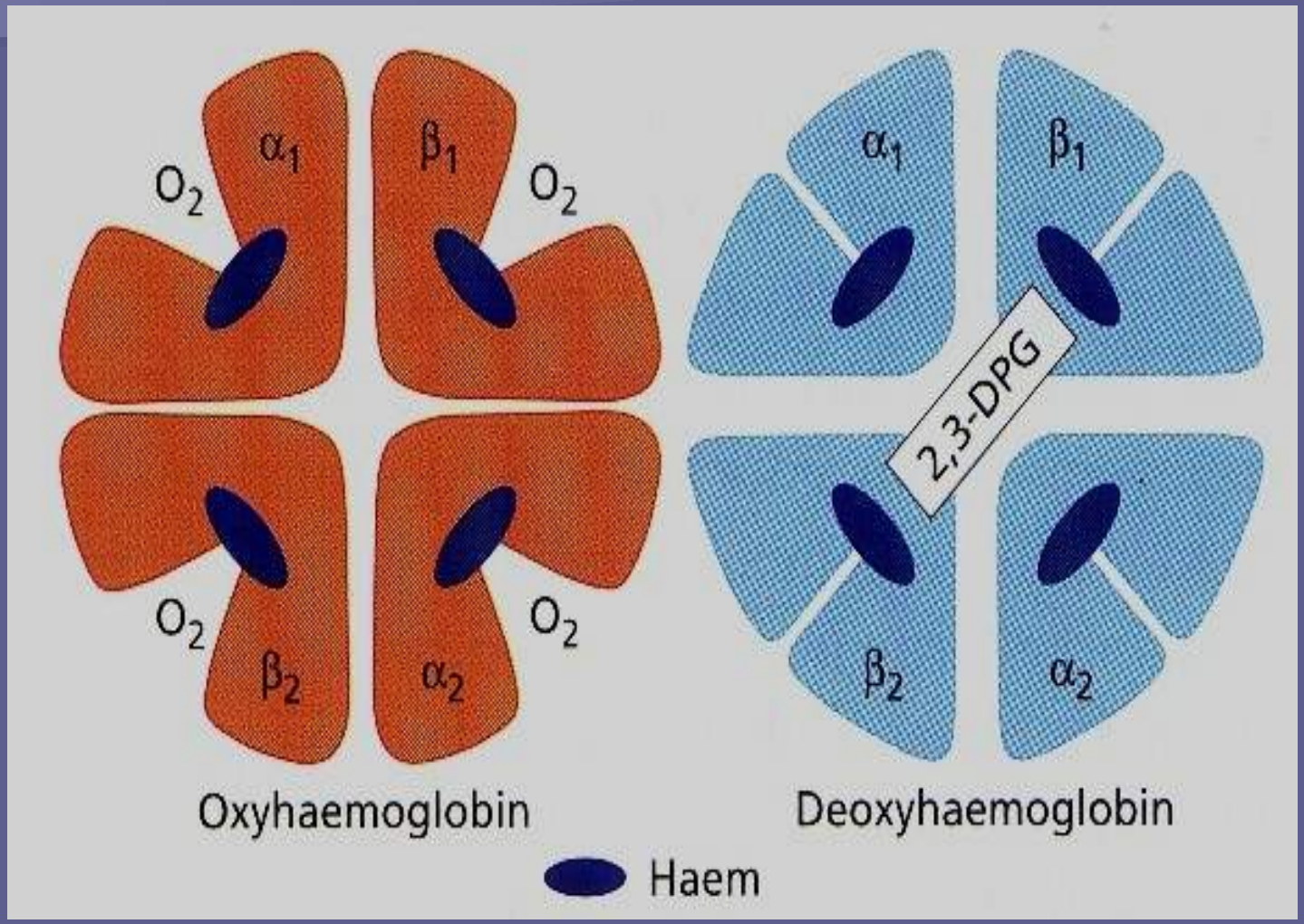
Adult haemoblobin

	Hb A	Hb A_2	Hb F
structure	$\alpha_2\beta_2$	$\alpha_2\delta_2$	$\alpha_2\gamma_2$
Normal %	96-98 %	1.5-3.2 %	0.5-0.8 %

Functions of Haemoglobin

- Oxygen delivery to the tissues
- Reaction of Hb & oxygen
  - Oxygenation not oxidation
  - One Hb can bind to four O_2 molecules
  - Less than .01 sec required for oxygenation
  - β chain move closer when oxygenated
  - When oxygenated 2,3-DPG is pushed out
  - β chains are pulled apart when O_2 is unloaded, permitting entry of 2,3-DPG resulting in lower affinity of O_2

Oxy & deoxyhaemoglobin



Summary

- Normal structure including the proportion of globin chains are necessary for the normal function of haemoglobin
- Reduced haemoglobin in the red blood cells due to any abnormality of any of its constituents result into a clinical situation called anaemia
- Metabolic & other abnormalities result into abnormal oxygen supply to the tissue

Anemia

- Anemia is a reduction in the number of RBCs, the quantity of hemoglobin, or the volume of RBCs
- Because the main function of RBCs is oxygenation, anemia results in varying degrees of hypoxia

Prevalent conditions

- Blood loss
- Decreased production of erythrocytes
- Increased destruction of erythrocytes

Clinical Manifestations:

- 1. Pallor.
  2. Fatigue, weakness.
  3. Dyspnea.
  4. Palpitations, tachycardia.
  5. Headache, dizziness, and restlessness.
  6. Slowing of thought.
  7. Paresthesia.

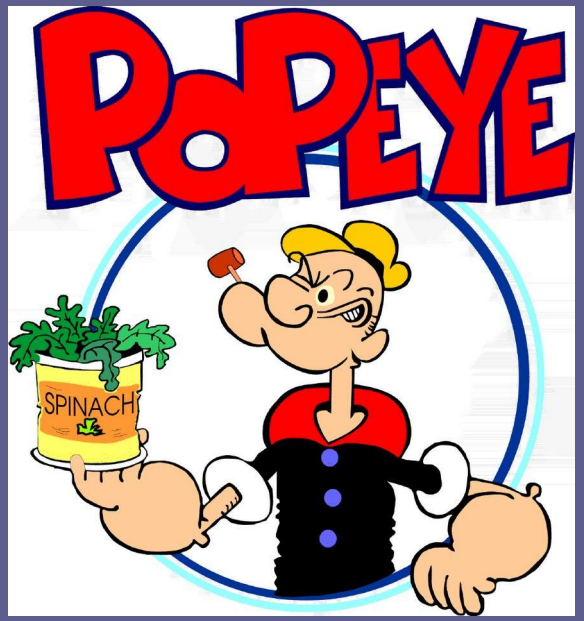
Anemia Caused by Decreased Erythrocyte Production

- Iron Deficiency Anemia
- Thalassemia
- Megablastic Anemia

Iron-Deficiency Anemia

Etiology

1. Inadequate dietary intake
   - Found in 30% of the world's population
2. Malabsorption
   - Absorbed in duodenum
   - GI surgery
3. Blood loss
   - 2 mls blood contain 1mg iron
   - GI, GU losses
4. Hemolysis



Clinical Manifestations

- Most common: pallor
- Second most common: inflammation of the tongue (glossistis)
- Cheilitis=inflammation/fissures of lips
- Sensitivity to cold
- Weakness and fatigue

Collaborative Care

- Treatment of underlying disease/problem
- Replacing iron
- Diet
- Drug Therapy
  - Iron replacement

Thalassemia

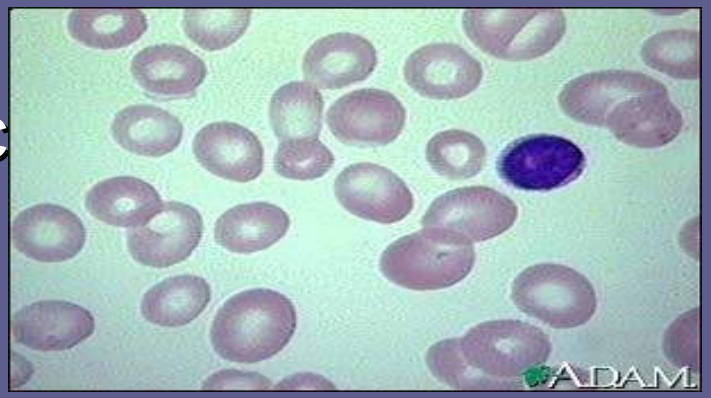
- Etiology
  - Autosomal recessive genetic disorder of inadequate production of normal hemoglobin
  - Found in Mediterranean ethnic groups
- Clinical Manifestations
  - Asymptomatic → major retardation → life threatening
  - Splenomegaly, hepatomegaly

Collaborative Care

- No specific drug or diet are effective in treating thalassemia
- Thalassemia minor
  - Body adapts to ↓ Hgb
- Thalassemia major
  - Blood transfusions with IV deferoxamine (used to remove excess iron from the body)

Megaloblastic Anemias

- Characterized by large RBCs which are fragile and easily destroyed
- Common forms of megaloblastic anemia
  1. Cobalamin deficienc
  2. Folic acid deficienc



This picture shows large, dense, oversized, red blood cells (RBCs) that are seen in megaloblastic anemia.

Cobalamin (Vitamin B_{12}) Deficiency

- Cobalamin Deficiency--formerly known as pernicious anemia
- ***Vitamin B_{12}*** *(cobalamin) is an important water-soluble vitamin.*
- *Intrinsic factor* (IF) is required for cobalamin absorption
- Causes of cobalamin deficiency
  - Gastric mucosa not secreting IF
  - GI surgery →loss of IF-secreting gastric mucosal cells
  - Long-term use of H_2-histamine receptor blockers cause atrophy or loss of gastric mucosa.
  - Nutritional deficiency
  - Hereditary defects of cobalamine utilization

Clinical manifestations

- General symptoms of anemia
- Sore tongue
- Anorexia
- Weakness
- Parathesias of the feet and hands
- Altered thought processes
  - Confusion → dementia

Collaborative Care

- Parenteral administration of cobalamin
- ↑ Dietary cobalamin does not correct the anemia
  - Still important to emphasize adequate dietary intake
- Intranasal form of cyanocobalamin (Nascobal) is available
- High dose oral cobalamin and SL cobalamin can use be used

Folic Acid Deficiency

- Folic Acid Deficiency also causes megablastic anemia (RBCs that are large and fewer in number)
- Folic Acid required for RBC formation and maturation
- Causes
  - Poor dietary intake
  - Malabsorption syndromes
  - Drugs that inhibit absorption
  - Alcohol abuse
  - Hemodialysis

Clinical manifestations

- Clinical manifestations are similar to those of cobalamin deficiency
- Insidious onset: progress slowly
- Absence of neurologic problems
- Treated by folate replacement therapy
- Encourage patient to eat foods with large amounts of folic acid
    - Leafy green vegetables
    - Liver
    - Mushrooms
    - Oatmeal
    - Peanut butter
    - Red beans

Aplastic Anemia

- Characterized by Pancytopenia
  - ↓ of all blood cell types
    - RBCs
    - White blood cells (WBCs)
    - Platelets
  - Hypocellular bone marrow
- Etiology
  - Congenital
    - Chromosomal alterations
  - Acquired
    - Results from exposure to ionizing radiation, chemical agents, viral and bacterial infections

Clinical Manifestations

- Gradual development
- Symptoms caused by suppression of any or all bone marrow elements
- General manifestations of anemia
  - Fatigue
  - Dyspnea
  - Pale skin
  - Frequent or prolonged infections
  - Unexplained or easy bruising
  - Nosebleed and bleeding gums
  - Prolonged bleeding from cuts
  - Dizziness
  - headache

Treatment

- Identifying cause
- Blood transfusions
- Antibiotics
- Immunosuppressants
- Bone marrow transplantation

Anemia Caused By Blood Loss

- Acute Blood Loss
- Chronic Blood Loss
- Result of sudden hemorrhage
  - Trauma, surgery, vascular disruption
- Collaborative Care

1. Replacing blood volume
2. Identifying source of hemorrhage
3. Stopping blood loss

Anemia caused by Increased Erythrocyte Destruction

- Hemolytic Anemia
  - Sickle Cell disease
  - Acquired Hemolytic Anemia
  - Hemochromatosis
  - Polycythemia